Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	14
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2541.77	3013	1427.44	px^2
major_axis_length_px	lacuna shape	108.237	124.464	40.2415	px
minor_axis_length_px	lacuna shape	31.7238	33.1799	10.2747	px
aspect_ratio	lacuna shape	3.5771	3.6285	1.3089	unitless
eccentricity	lacuna shape	0.935	0.9613	0.0596	unitless
solidity	lacuna shape	0.7472	0.7672	0.1322	unitless
roots_count	headline	11	10	5.8045	count
ring_length_r30_px	headline	433.143	399	196.829	px
ring_length_r60_px	reported	995.929	923.5	437.73	px
owned_length_px	ownership-dependent	2104.03	1924.3	1562.1	px
edge_count	ownership-dependent	92.2857	78.5	70.267	count
mean_edge_length_px	ownership-dependent	22.7907	22.6947	4.1278	px
perimeter_px	normalised	280.602	305.986	127.981	px
ring_area_r30_px2	normalised	10209.4	11138.5	3432.15	px^2
ring_area_r60_px2	normalised	23888.4	25904.5	7904.56	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9914	1	0.0263	unitless
ring_density_r30	normalised	0.0414009	0.0406485	0.00716751	px^-1
ring_density_r60	normalised	0.0404847	0.039246	0.00640785	px^-1
roots_per_100px_perimeter	normalised	4.0757	4.0826	1.2805	per 100 px
ring_attached_length_r30_px	network v2	379.286	335	202.625	px
ring_attached_length_r60_px	network v2	789	619	478.954	px
ring_length_w_r30_px	network v2	492.185	450.954	225.176	px
ring_length_w_r60_px	network v2	1128.49	1034.47	500.315	px
sholl_crossings_r10	network v2	9.3571	9	5.078	count
sholl_crossings_r20	network v2	13.1429	13	5.4187	count
sholl_crossings_r30	network v2	13.2143	12.5	5.7536	count
ring_width_mean_r30_px	width	7.4897	7.4857	0.4228	px

Per field
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measure	value	units
analysed_area_px2	1.00885e+06	px^2
total_skeleton_length_px	39263	px
canalicular_length_density_per_px	0.0389187	px^-1
junction_count	863	count
junction_density_per_px2	0.00085543	px^-2
lacuna_count	17	count
lacunae_per_px2	1.685e-05	px^-2
skeleton_component_count	248	count

mean_component_length_px	158.319	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0440846	px^-1
field_density_without_flagged_per_px	0.0388656	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	362	px
unattached_component_count	19	count
canalicular_length_density_attached_per_px	0.0389187	px^-1
field_length_density_with_unattached_per_px	0.0392775	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	8.1545	px
width_median_px	8.2462	px
width_p10_px	5.6569	px
width_p90_px	10.7703	px
width_pixels_used	38017	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	556	44.867	25.9744	1.7274	0.8154	0.5265	-0.4071	66.9281	106.103	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True
17								368.378	576.748	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	218	581	894.411	31	28.852	7.8557
2	6	194	502	825.027	30	27.5009	7.7664
3	7	281	689	1352.28	46	29.3975	7.1332
4	6	255	657	344.522	19	18.1327	7.3342
5	7	380	1031	2055.6	84	24.4714	7.2803
6	22	784	1661	5837.1	238	24.5256	7.6259
7	14	410	749	822.198	36	22.8388	7.4551

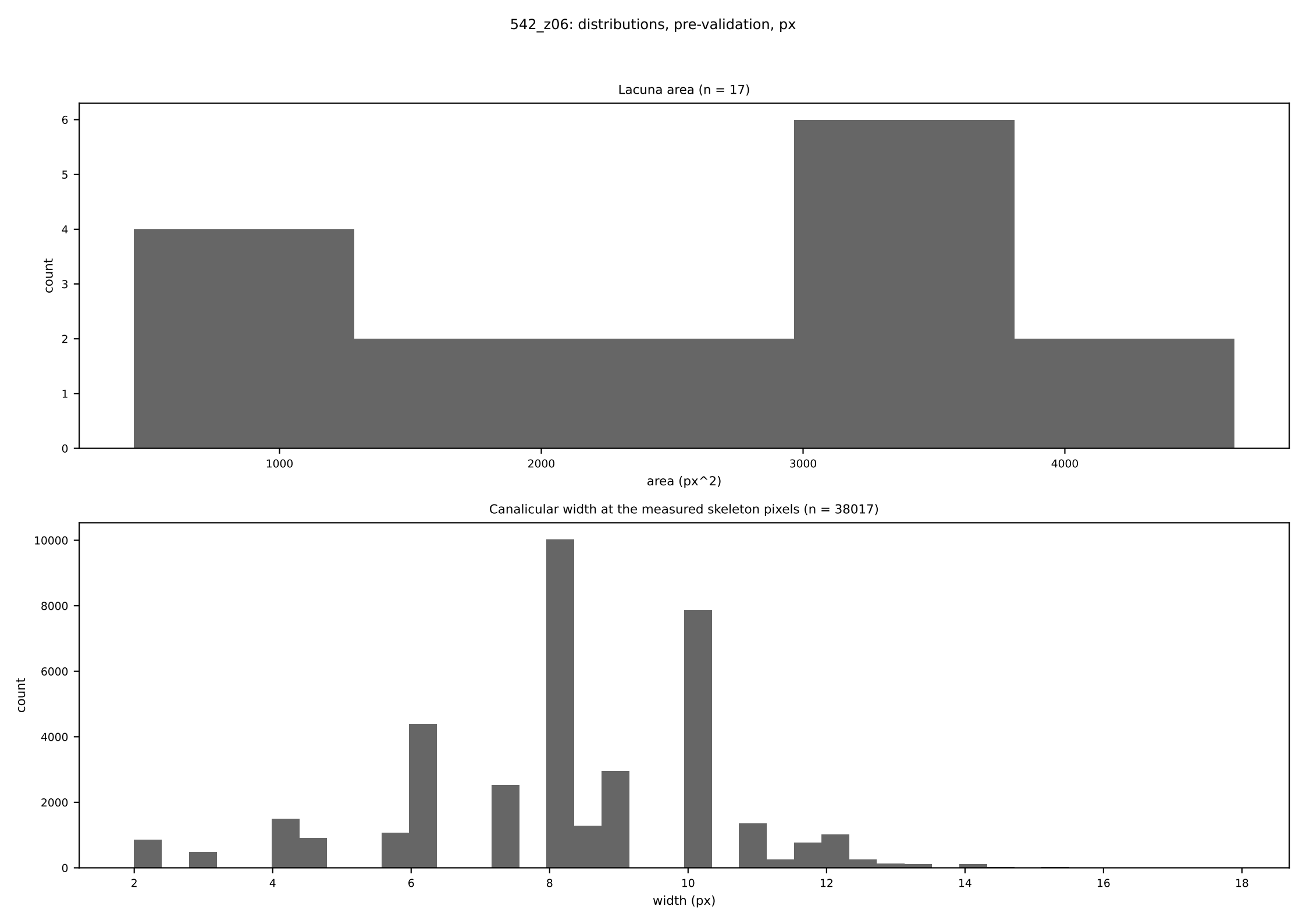
8	14	459	887	1793.01	83	21.6025	7.5163
9	6	388	960	2255.05	100	22.5506	8.5387
10	16	465	1270	3369.32	138	24.4154	7.7735
11	18	767	1786	4279.83	224	19.1064	7.4169
12	13	601	1342	1454.21	71	20.4819	7.1869
13	16	608	1318	2719.25	135	20.1426	7.6692
14	6	318	855	2139	74	28.9054	6.6409
15	5	274	679	862.997	44	19.6136	7.6559
16	3	143	494	63.4558	6	10.576	5.6111
17	3	154	236	209.983	14	14.9988	7.5189

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0388107	0.0457697	2.7877
2	126.468	6226	14794	1	1	0.0311597	0.0339327	4.7443
3	160.64	6976	18799	1	0.9774	0.040281	0.0366509	4.3576
4	151.782	6752	15321	1	1	0.0377666	0.0428823	3.953
5	322.043	11663	28685	1	1	0.0325817	0.0359421	2.1736
6	438.517	13901	31616	1	1	0.0563988	0.0525367	5.0169
7	211.924	8423	18846	1	1	0.0486762	0.0397432	6.6061
8	359.191	12383	27034	1	0.9025	0.037067	0.0328105	3.8976
9	302.451	10940	24775	1	1	0.0354662	0.0387487	1.9838
10	309.522	11337	28286	1	1	0.0410161	0.0448985	5.1693
11	468.274	15363	34096	1	1	0.0499251	0.0523815	3.8439
12	422.978	13165	30001	1	1	0.0456514	0.0447318	3.0734
13	379.848	13296	32108	1	1	0.045728	0.041049	4.2122
14	217.924	8830	22986	1	1	0.0360136	0.0371966	2.7533
15	332.894	10505	23806	0.8602	0.7919	0.0260828	0.0285222	1.502
16	205.776	6766	16357	0.7841	0.714	0.0211351	0.0302011	1.4579
17	56.8701	3677	7091	1	1	0.041882	0.0332816	5.2752

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	574	251.066	663.701	6	5	5
2	157	376	224.137	565.517	4	8	10
3	210	367	327.877	786.896	5	8	5
4	211	338	285.966	736.257	5	7	8
5	319	837	437.647	1179.53	8	14	13
6	781	1566	902.465	1908.83	20	20	22
7	351	547	464.262	849.512	8	12	11
8	415	691	514.304	993.666	10	11	10
9	301	537	436.09	1075.26	11	14	12
10	444	1094	534.517	1452.87	10	14	15
11	686	1687	873.654	2015.19	14	22	25
12	530	1121	674.286	1515.8	14	16	15
13	570	1215	688.671	1491.87	15	22	19
14	227	546	358.907	969.495	5	11	14
15	187	280	305.522	756.813	7	9	14
16	108	108	158.539	546.132	4	7	6
17	108	124	167.811	258.166	2	5	6


542_z06: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
0
1
2
3
4
5
6
1000
2000
3000
4000
Canalicular width at the measured skeleton pixels (n = 38017)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18